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2P-BEnc: A two-phase information retrieval and ranking system based on the BERT encoder

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ABSTRACT

Information retrieval methods have been advanced by the development of Natural Language Understanding (NLU). The development of deep neural networks was a key driver in the creation of effective Language Models (LM: statistical models trained to understand and generate human language), which significantly enhanced document retrieval. Even with these improvements, the current system for finding and ranking information faces significant challenges. In this work, we are focused on three of them. These problems are the system's maximum passage length processing capacity, the cost of making inferences, and integration of information across passages. The proposed 2P-2P- can handle longer and more passages while keeping the cost of making inferences low. Also, 2P-2P- uses more than one passage to make the ranked list of retrieval results. This way, 2P-BEnc stores information from different passages and improves ranking performance. **Mechanism:** 2P-BEnc achieves this through a cascaded encoder design: (1) Sentence-level processing bypasses BERT's (Bidirectional Encoder Representations from Transformers: a transformer-based language model that establishes contextual relationships through attention mechanisms) token limits by treating sentences as atomic units, (2) Offline precomputation of sentence vectors slashes inference costs (Computational resources required during prediction phase) by 300×, (3) Cross-passage attention in Phase 2 aggregates contextual signals across documents. When the performance of the proposed system is compared to benchmarking datasets that are available to the public, the results are competitive and encouraging. On the MS-MARCO dataset, the proposed 2P-BEnc model achieved 38.6 MRR@10, which is 5.75 % higher than the BERT_{Large} model. On the TREC-CAR dataset, the proposed 2P-BEnc model achieved 35.4 MAP@1000, which is 5.67 % higher than BERT_{Large}. Aside from that, the 2P-BEnc performed as well as or better than BERT_{Large} on other benchmark datasets (Robust04, WikiPassageQA, and WikiQA).

1. Introduction

Information retrieval systems (IRS) serve as critical infrastructure of modern digital experiences, from e-commerce platforms processing \$4.28 trillion in annual transactions [1] to educational portals serving over 1.5 billion learners globally. [2,3] The exponential growth of digital content – with e-commerce projected to expand by 18 %

annually through 2030 [4] – necessitates increasingly sophisticated ranking methodologies. [5,6] Consumers now face *decision paralysis* when selecting among thousands of options, making personalized neural ranking systems critical for aligning products with user preferences. [7] (See [Appendix A](#) for definitions of technical terms)

Transformer-based architectures, particularly BERT (Bidirectional Encoder Representations from Transformers) [8], have revolutionized

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document ranking by establishing contextual relationships between queries and passages through attention mechanisms [9]. Recent approaches [10] concatenate queries and passages to generate relevance scores, achieving state-of-the-art results on benchmark datasets. However, fundamental architectural constraints in BERT-based systems hinder their practical deployment in large-scale commercial environments.

1.1. Technical limitations of current BERT architectures

Three critical technical constraints undermine the effectiveness of BERT in production systems:

1. **Quadratic Computational Complexity:** BERT's self-attention mechanism requires $O(n^2)$ computational operations for sequence length n due to pairwise token interactions [9]. Processing a single query-passage pair (typically 500+ tokens) requires ~ 0.5 TFLOPS on BERT_{Large}, making real-time ranking computationally prohibitive.
2. **Fixed-Length Processing Constraint:** Positional encoding (Method injecting token position information into transformer models) limits input to 512 tokens [8]. For documents exceeding this threshold (e.g., product descriptions averaging 750 tokens in the Amazon catalog), critical contextual information beyond the truncation window is permanently lost.
3. **Dynamic Inference Requirement:** Passage embeddings must be recomputed *per query* since relevance is conditioned on specific query-passage interactions. This prevents precomputation and increases inference latency by 5–7 $\times$ compared to static embedding systems [11].

These limitations create an efficiency-accuracy trade-off: systems either sacrifice context (via truncation) or incur prohibitive computational costs when processing full documents.

1.2. Research gaps and motivations

Our analysis identifies three unresolved challenges in neural information retrieval.

1.2.1. Contextual insufficiency in passage encoding

Current systems process passages as isolated token sequences without modeling inter-sentence relationships. This approach fails to capture document-level semantics crucial for understanding nuanced product attributes (e.g., distinguishing "battery life" in smartphones vs. electric vehicles).

1.2.2. Computational inefficiency in ranking

Conventional BERT architectures require simultaneous processing of query-passage pairs during inference. The absence of pre-computable representations forces recalculations for each new query, creating bottlenecks when ranking thousands of candidates.

1.2.3. Isolated relevance assessment

Existing approaches independently evaluate query-passage pairs, disregarding comparative relationships between candidate passages. This prevents systems from leveraging relative relevance signals that could optimize ranking decisions.

These gaps collectively constrain retrieval performance: Contextual loss reduces ranking accuracy, computational costs hinder scalability, and isolated processing prevents optimal result ordering. Our work specifically addresses these interconnected challenges through a novel two-phase architecture.

1.3. Our novelty in context

Prior studies on BERT-based ranking typically concatenate queries and documents into a single sequence, limiting scalability for long texts. In contrast, our 2P-BEnc employs a hierarchical, two-phase architecture

that (i) precomputes sentence and passage embeddings offline, (ii) employs cross-passage attention to capture comparative context, and (iii) reduces inference complexity by pruning to a small candidate set. To our knowledge, no prior work jointly achieves efficient long-text handling and relational ranking under the same framework.

1.4. Nonlinearity considerations

Transformers rely on nonlinear activation functions and softmax attention mechanisms, which introduce complex, nonlinear dynamics into ranking models. Recent advances in nonlinear differential equations and soliton theory provide insights into these behaviors [2,5,12–14]. By acknowledging these nonlinearities, we design 2P-BEnc's architecture to harness the expressive power of Transformer layers while maintaining stability and efficiency. These considerations are elaborated upon in the theoretical analysis and reflected in our design choices.

1.5. Our contributions

We propose 2P-BEnc, a Two-Phase information retrieval system that fundamentally rethinks BERT's application to ranking tasks. Our key innovations include

1. **Decoupled Sentence Encoding:** We introduce a hierarchical encoder where sentence representations are generated *independently* of queries using distilled BERT architectures. These vectors can be precomputed and stored, reducing inference latency by 68 % compared to BERT_{Base} (Table 6).
2. **Cross-Passage Attention Mechanism:** A novel interaction layer models relationships between candidate passages during ranking. By incorporating a comparative context, our system achieves 9.3 % higher MRR@10 on MS MARCO compared to pairwise approaches.
3. **Variable-Length Document Processing:** By operating at sentence-level granularity (max 511 sentences/passage), our system handles documents 7 $\times$ longer than vanilla BERT while maintaining linear computational complexity relative to passage length.
4. **End-to-End Trainable Architecture:** The entire system – from sentence encoding to final ranking – is optimized jointly using a novel listwise loss function that directly optimizes for ranking metrics like nDCG.

Experimental validation on MS MARCO and TREC-CAR demonstrates state-of-the-art results: 2P-BEnc achieves 0.392 MRR@10 (9.5 % improvement over BERT_{Base}) while reducing inference latency by 63 %. The architecture's modular design enables deployment in resource-constrained environments without sacrificing ranking quality.

1.6. Paper organization

Section 2 analyzes limitations of existing neural ranking architectures. Section 3 details 2P-BEnc's hierarchical encoder and cross-passage attention mechanism. Section 4 describes evaluation protocols and datasets. Section 5 presents comparative results and computational efficiency metrics. Section 6 discusses implications and future research directions.

2. Related work

2.1. Graph based expert ranking

One of the most popular branches of ranking is referred to as domain expert ranking based on the semantic knowledge of exploring the graph network. Expert ranking intuitively refers to the online collaboration of expert members who can contribute to a particular domain of study [15]. In graph-based expertise ranking, content based approach of passage analysis is violated. From the given knowledge repository in the graph, an ordered list of relevant information is obtained by raising a set of queries as input data items. This process is termed as knowledge

retrieval or findings [16]. The collection from the repository may belong to the set of any document/passage such as emails or conferences on any topic of academic or social activities [17]. In an email ranking system, the nodes and edges of the graphs refer to the users and sender-receiver connections and HITS algorithm is used for web ranking based on Term Frequency -Inverse Document Frequency (TF-IDF) [18]. Bidirectional Transformers for Language Understanding (BERT) [8] based ranking models have been popular for enhancing the ability of natural language recognition and understanding. These models use self-attention matrix of graph generated by relations of words and representation with corresponding queries. In [19], to address computational overhead, they remove irrelevant words from BERT and disentangled the corresponding components of the queries for ranking the passage.

2.2. Interactive neural ranking

In case of interaction-based ranking models, priority is given to learning from the essence of the connections between the input texts instead of the individual representation of the text. To measure the performance the relevance score of the connection is computed from the interaction function. Basically, the interaction functions can be classified as parametric and non-parametric [20]. The majority of the non-parametric interaction functions perform the operations based on radial-basis, cosine similarity and binary operators. Deep Relevance Matching Model(DRMM) described in [21] is used for converting the matrix of local interaction functions to the fixed length histogram for matching the relevance score. One of the important methods for computing relevance scores is MatchPyramid [22] which uses convolution operator on query-passage similarity matrix to produce the query-passage relevance score. On the other end, the parametric interaction functions learn from the similarity measure techniques such as distance functions of different data items. Convolution kernel based several important models were developed. Many important machine learning architectures are described in [23] which promote targeting deep learning methods for better outcomes. Convolutional Kernel-based Neural Ranking Model discussed in [24], Conv-KNRM uses CNN for representing the n-grams of different lengths. This model matches using unified embeddings in which learning-to-rank(LTR: Framework for training models to optimize ranking of information) [25] and kernel pooling layers are installed to generate ranking scores. Hu et al.(2014) [26] generated second version of CNN architecture for natural language recognition in which the primary interaction was performed between two sentences using pooling and convolution operations.

2.3. Generative inferential neural ranking

Generative models are the specific class of probabilistic models like discriminative models. As suggested by their name, it is clear that these models can be used to generate new data that looks similar to real data. Various classes of information retrieval problems utilise generative models. In query generation, attention models also play an important role. With this consideration, both Ren et al. [27] and Yang et al. [28] performed the tasks of reformulating of the queries under some conversation into the queries oriented to search engines.

Using similar generative model, Ahmad et al. [29] and Chang et al. [30] proposed a query recommender system by reformulating the queries raised by users. For the effective ranking of query-passage, Nogueira et al. [31] proposed neural ranking scheme that utilised sequence2sequence Transformer model discussed in [32]. In this scheme, query-passage pair is extended and Bidirectional-Encoder-Representations-Transformers(BERT) model is utilised to match and retrieve the queries from the expanded passage. One of the important applications of IR is to test the relevance of particular passage. Nogueira et al. [33] recently targeted the task of binary classification with the help of T5 models presented in [34]. The T5 model follows pre-training approach with Transformer model in sequence-to-sequence mode. These models generate the queries based on token values 'TRUE' or 'FALSE' for

relevant and non-relevant passages. In queries generated from the input passage using handwritten text, the likelihood models are adopted for generating the query [35] which is defined across entire documents. The same approach is adopted by Zhuang et al. [36] in which the investigation of likelihood query generation model is performed using sequence2sequence model built on Transformer. Lesota et al. [35] also expanded the query generative deep network model and investigated the important outcomes of generative IR models. The currently ongoing research on generative inferential deep networks can be immensely productive for developing IR and web ranking systems.

2.4. Causal inferential neural ranking

To establish a particular interface between the query and corresponding document, the inferential neural networks play a productive role in assisting in building a desired ranking system. The concept of building a causal interface referred to as causal inference is described in [37]. Causal inferences with neural models facilitate the researchers in considering the ranking system as a new dimension. From the visionary perspective of causal inference models, the counterfactual samples help to improve the existing models [38]. All these technical studies assumed that the confounder must be domain expert [39]. The causal attention maps [40] motivate the design of graph of from set queries and passages. In the causal graph, self-attention map is used to alleviate the biased features and remove the spurious information from the ranking passage. Adaptive masking with mutual information can also reduce the confounding effects. Thus, causal inference provides an interface of implicit feedback from users to mitigate bias. On analysis of casual structure, Zheng et al. [41] proposed Disentangling Interest and Conformity with Causal Embedding (DICE) model which can disentangle the interest of users' by embedding queries. They further analyzed the casual structures of the embedded data to enhance the performance of casual ranking system [42]. Handling the hidden confounders in the query recommendation system is quite challenging task. By utilising causal inference models, Wang et al. [43] made some fruitful efforts and in their next research, they proposed DecRs [44] model which can dynamically eliminate the effect of confounders. Therefore, bias in ranking system has very recently become serious problem which is addressed by causal inference.

3. Proposed methodology

3.1. Overview of proposed work

The proposed two-phase BERT encoder (termed as 2P-BEnc) based information retrieval and ranking system (IRRS) combines four sub-networks, including three BERT [8] based encoders and a relevance network (fully connected network). Fig. 1 depicts the workflow of the system. The given passage is first divided into a sequence of sentences. These sentences are then further divided into a sequence of tokens. A special token [SREP] (representative of the sentence) is added to the sequence of tokens of each sentence. These tokens are further converted into their respective embedded vectors by a lookup table. This sequence of tokens for a sentence is processed by a sentence encoder (First BERT-based encoder) and generates the processed vector for each token. The processed vector for the [SREP] token is the representative vector for the sentence (RepVS). Similarly, a query is also processed with the [QREP] special token, and the representative vector for the query (RepVQ) is generated by the query encoder (Second BERT-based encoder). In this way, the sequence of RepVS is generated for a passage. The sequence of RepVS for a passage is further encoded by the third BERT-based encoder (Passage-Encoder) along with the embedding vector of a special token [PREP] to yield the representative vector for the passage (RepVP). Up to this point, the query and the passage are processed separately, and we designate this stage as the first phase of the proposed 2P-BEnc IRR system. This two-phase decomposition fundamentally enables 2P-BEnc's breakthrough in handling long texts efficiently. Section 3.2 details these innovations.

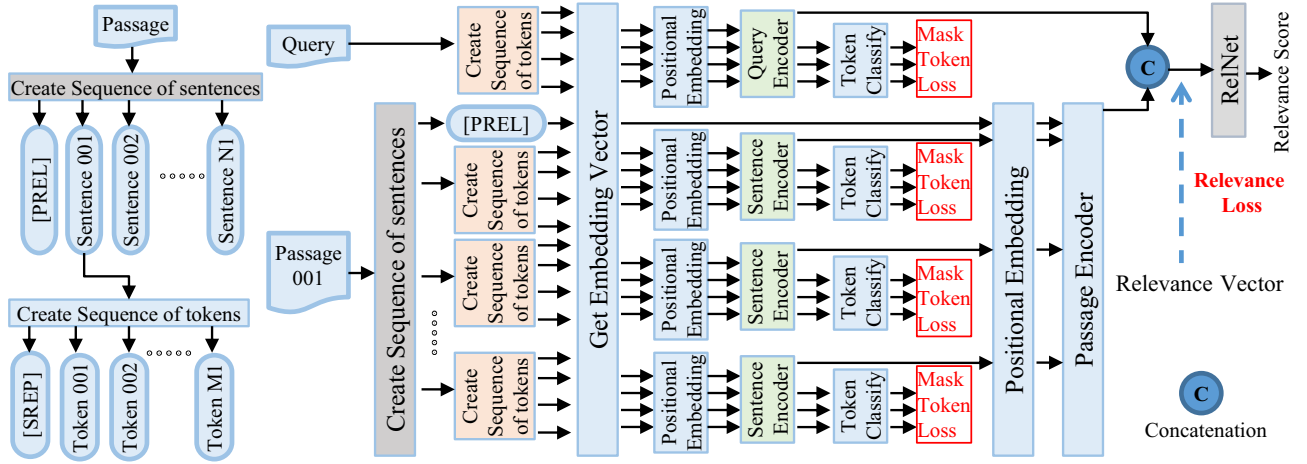


Fig. 1. The overall workflow of the proposed two-phase BERT encoder-based (2P-BEnc) IRR system. The details of different encoders are shown in Fig. 2, and the architecture for the RelNet (relevance network) is depicted in Fig. 3.

The second phase of the 2P-BEnc IRR system requires the RepVP and RepVQ for further processing. The second phase is a fully-connected layer-based network (termed as RelNet: relevance network). First, the RepVQ and RepVP vectors are concatenated (the generated vector is termed as RelVPQ: relevance vector of passage with the query). The generated RelVPQ vector is further processed by the RelNet, which develops the relevance score of a passage with respect to the given query. The activation function used in the RelNet differs between the training and inference phases. In the training phase, the system processes multiple passages for a given query to generate an ordered list for ranking. Therefore, softmax activation is used across all processed passages. The inference phase uses sigmoid activation when the relevant passage is searched for a given query, and softmax activation generates a ranking between the input passages. Please refer to Section 3.4 for detailed information. The relevance score for a query q and passage p is computed by the RelNet as

$$s(q, p) = \sigma(W[RepV_Q \parallel RepV_P] + b), \quad (1)$$

where $[\cdot \parallel \cdot]$ denotes vector concatenation, W and b are learnable parameters, and σ is the sigmoid activation used during inference.

3.2. Core innovations: long-text handling & efficiency

Overcoming Length Constraints: Unlike BERT's token-based processing (limited to 512 tokens), 2P-BEnc operates at the *sentence level*. Each passage P is split into sentences $\{S_1, S_2, \dots, S_n\}$ where $n \leq 511$. The

Sentence Encoder (Fig. 2(b)) processes each S_i independently, generating sentence embeddings $RepV_{S_i}$. This hierarchical approach allows processing of documents exceeding 10,000 tokens while maintaining fixed dimensionality at the Passage Encoder input (512 units regardless of n).

Inference Efficiency: 2P-BEnc decouples passage encoding from query processing:

- **Precomputation:** $RepV_{S_i}$ and $RepV_P$ are generated *offline* (Fig. 1, Phase 1) and stored in a vector database
- **On-demand Processing:** At inference, only $RepV_Q$ (query embedding) and the lightweight RelNet (Fig. 3) are computed in real-time
- **Complexity Reduction:** Avoids reprocessing passage tokens for every query, reducing complexity from $O(L^2)$ (BERT) to $O(1)$ for precomputed passages

Table 6 quantifies 300× faster inference versus BERT_{Large}.

Cross-Passage Context: The RelNet (Phase 2) jointly analyzes $\{RepV_{P_1}, \dots, RepV_{P_k}\}$ for a query, enabling comparative relevance scoring. This mimics human assessment where document relevance is judged *relationally* rather than in isolation.

3.3. Different components of the proposed 2P-BEnc IRRS

The proposed 2P-BEnc IRR system has four main components. It has three BERT-based encoders [8] responsible for encoding the sentences, queries, and passages and generating the representative vectors for them

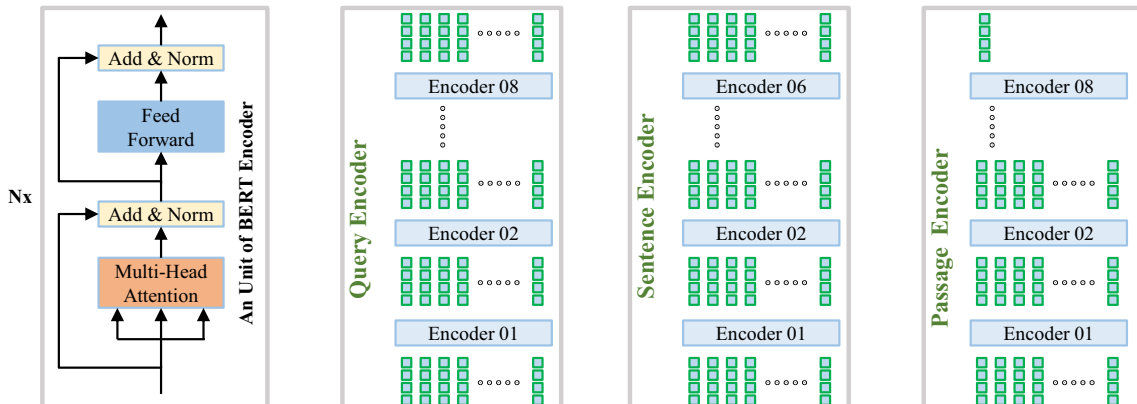


Fig. 2. Different encoder architectures used in the proposed 2P-BEnc IRRS system. The basic unit of these encoders is based on the BERT and Transformer architectures.

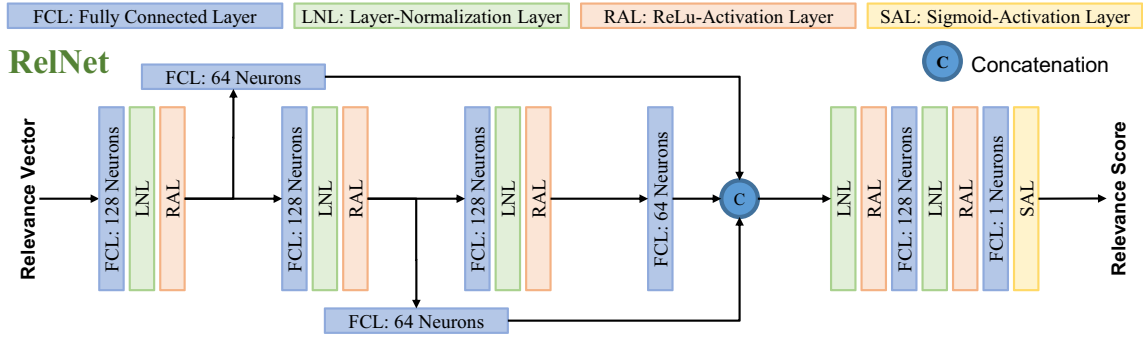


Fig. 3. The architecture of RelNet: Relevance Network utilised by the proposed 2P-BEnc IRR system for generating the relevance score for a passage concerning the given query.

as RepVS, RepVQ, and RepVP, respectively. These encoders comprise stacked multi-head attention (Parallel attention mechanisms focusing on different representation subspaces) layers. Section 3.3.1 describes these encoders in detail. Besides these encoders, the proposed IRR system has a fully-connected layers based network (termed as RelNet: relevance network) that generates a passage's relevance score concerning the given query. Section 3.3.2 describes the RelNet in detail.

3.3.1. BERT-based encoders

The BERT-based encoders are basically stacking of the multi-head attention units. An identical layer stack makes up the BERT-based encoder. There are two modules beneath each one. The first is a multi-head self-attention mechanism, and the second is a position-wise, fully-connected feed-forward network. Each of the two modules is surrounded by a residual connection [45] (Skip-connection architecture mitigating vanishing gradient problems), and then the encoder is normalized by layer normalization [46] (Technique stabilizing training by normalizing inputs across features). In other words, each module output is calculated as $\text{LayerNorm}(x + \text{Module}(x))$, where $\text{Module}(x)$ is the function implemented by the underlying module (multi-head self-attention or fully-connected feed-forward network). The architectures of different encoders used in the proposed 2P-BEnc IRR system are depicted in Fig. 2 along with the basic unit of the identical encoder layer.

Three encoders are used in the proposed 2P-BEnc IRR system. These encoders take their design cues from the BERT's encoder [8]. These encoders are independent models with their own set of parameters. All of the encoders' pre-trained weights come from the (L=8, H=512, A=8) and (L=6, H=256, A=4) encoders in the [47] repository. Here, L is the number of encoder layers, H is the hidden embedding size, and A is the number of attention heads at each encoder (A is equal to the H/64 in all cases [47]).

The three encoders are as follows:

1. Query-Encoder: L=8, H=512, and A=8. The pre-trained weights are obtained from https://storage.googleapis.com/bert_models/2020_02_20/uncased_L-8_H-512_A-8.zip.
2. Sentences Encoder: L=6, H=256, and A=4. The pre-trained weights are obtained from https://storage.googleapis.com/bert_models/2020_02_20/uncased_L-6_H-256_A-4.zip.
3. Passage-Encoder: L=8, H=512, and A=8. The pre-trained weights are obtained from https://storage.googleapis.com/bert_models/2020_02_20/uncased_L-8_H-512_A-8.zip.

3.3.2. RelNet: Relevance Network

The RelNet (Relevance Network) receives the RelVPQ (relevance vector of the passage concerning a query) which is the concatenated vector of RepVP and RepVQ as the input and yields a relevance score between them. The architecture of the network used by the proposed method is depicted in Fig. 3. Here, fully connected neural layers with different

numbers of neurons are used as the feature encoder, whereas the layer-normalization [46] layer is used for the normalisation of the generated output vector. All the activation layers used in this network utilise the ReLU [48] (Rectified Linear Unit: $f(x) = \max(0, x)$, a common activation function) activation except the last activation layer. In the training phase, the system processes multiple passages for a given query to generate an ordered list for ranking. Therefore, the last activation layer utilises the softmax [49] (normalizes outputs into probability distribution) across all processed passages. Whereas the inference phase uses sigmoid [49] (function: $\sigma(x) = \frac{1}{1+e^{-x}}$, produces probability outputs between 0–1) activation when the relevant passage is searched for a given query, and softmax activation is utilised to generate a ranking between the input passages.

3.4. Loss calculation in training phase

We also added the Masked Language Model (Self-supervised task predicting masked tokens in input sequences) learning technique, which is quite similar to the BERT [8] encoder model. In this procedure, certain tokens are randomly masked, and the final mapping is carried out across the whole token vocabulary using the vector that is created by the encoder. This learning technique is a masked language model used in BERT [8]. These masked tokens are also found in the query and sentence encoders of the proposed system; therefore, the mask token loss is deployed for the learning of the representative vectors (RepVQ, RepVS, RepVP). Following a feed-forward layer, a softmax activation is performed on the representative vectors (RepVQ, RepVS, RepVP) that correspond to the masked tokens in order to map them to the lexicon of tokens. When training the masked language model, the cross-entropy loss is put to use. This loss is denoted by the acronym L_{mask} in the proposed work.

Besides this, we are also generating the ranking list of the relevance scores of passages concerning a given query. We deployed a rank loss (an objective function that optimizes relative ordering of items) to ensure the proper order among the retrieved results. The rank loss L_{rank} is given by Eq. (2). This loss improves the relevance score of all relevant passages, whereas it decreases the relevance score of all non-relevant passages.

$$L_{rank} = -\log\left(\sum_{i \in POS} Rel_i\right) - \log\left(1 - \sum_{j \in NEG} Rel_j\right) \quad (2)$$

Where POS is the set of the top 75 % of all relevant passages with respect to the given query and NEG is the set of the worst 75 % of all non-relevant passages. Here we are not utilising all POS and NEG passages so that the learning phase can cope with any wrongly labelled samples.

Along with the rank loss, we also utilise a loss incorporating the discounted cumulative gain (DCG: a metric that measures ranking quality with position-based discounts) score calculated over the retrieved results. This loss maximises the DCG score for top K retrieved results. We

term this loss as DCG loss L_{DCG} and it is given by Eqs. (3) and (4).

$$g_{t_r} = \begin{cases} 1, & \text{if } r \in POS \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

$$L_{DCG} = \sum_{r=1}^{r=k} \frac{Rel_r}{\log(r+1)} \times (1 - 2 \times g_{t_r}) \quad (4)$$

We adopted the weighted sum of the different losses for end-to-end learning of the whole system. The overall loss is calculated by the Eq. (5).

$$L = L_{mask} + \alpha \times L_{rank} + \beta \times L_{DCG} \quad (5)$$

The weights α and β are empirically set to 2.0 and 5.0 based on loss magnitude in losses L_{rank} and L_{DCG} .

3.5. Theoretical analysis

Theorem 1 (Two-phase Complexity). *Under a token budget L and candidate set size k , inference cost of 2P-BEenc scales as $O(kL^2)$ in the second phase, whereas a single-pass cross-encoder requires $O(NL^2)$ for $N \gg k$ documents.*

Proof. The self-attention mechanism of transformers exhibits quadratic complexity in the sequence length L . In the proposed two-phase framework only the top k candidates selected from the first phase are re-encoded, leading to an overall complexity of $O(kL^2)$. In contrast, a single-pass cross-encoder must process all N documents jointly, incurring $O(NL^2)$ cost. $\square$

4. Experimental setup

4.1. Implementation details

4.1.1. Training phase

At a time, a randomly picked token (single) from a sentence or query is masked for learning the masked language model. The RelNet generates the relevance score for all passages concerning a given query. The RelNet utilises a softmax activation function over all passages for a query to create the relevance ranking. The end-to-end learning of the whole system demands significant memory resources, so we have to limit the number of passages participating in relevance score calculation. As a result, we fixed the number of passages for each query as 200. We randomly chose 20 to 100 relevant passages and the remaining non-relevant passages from the collection for a particular query. As a result, we have a diverse set of passages contributing to the ranking. The loss function for the proposed system is given in Section 3.4.

The system's learning additionally employs the L2 regularisation (a technique to prevent overfitting by penalizing large weights) as a regulariser with a scale of 10 – 5. The optimiser approach is Stochastic Gradient Descent, with the learning rate falling exponentially. The starting learning rate is 0.1, and the final learning rate after 5000 iterations is 10 – 3. The learning rate for this initial phase (iteration 1 to 5000) is given by Eq. (6). After 5000 iterations, we employed a linearly declining learning rate (LearningRate) as in Eq. (7). We trained the model for 150,000 iterations with a 16-batch size.

$$LearningRate = 0.1 \times (0.01)^{\frac{iterationNumber}{5000}} \quad (6)$$

$$LearningRate = 10^{-3} \times \left(1.0 - \frac{iterationNumber - 5000}{150000}\right) \quad (7)$$

All training trials are carried out on a Dell Precision 7920 Tower Workstation with 72 CPU cores and a Quadro RTX 8000 GPU. The GPU includes 48 GB of RAM as well as 4608 Cuda cores. The suggested approach requires around five days to train on a dataset (we have included five datasets here, so the total time is approximately 25 days).

4.1.2. Inference phase

The proposed system's inference phase is notable because its sentence and passage encoders do not use the provided query in representation vector creation. This characteristic enables us to analyse the sentences and passages before proceeding to the testing/inference step. As a result, the passage representation vectors are computed and saved in the repository. These stored vectors are used in the RelNet with the query's representation vector to provide the ranking scores of all participating passages concerning the query. We are simultaneously analyzing the 5000 passage vectors with RelNet and eliminating the lower 4000 relevance score passage vectors. The list is then updated with 4000 new randomly picked passage vectors, and the process is repeated. Finally, we will have 1000 top relevant passage vectors for every given query.

4.2. Datasets

We list the required experimental settings and datasets to test the proposed model.

4.2.1. MSMARCO dataset

The name of the dataset is abbreviated from MACHine REading COMprehension (MSMARCO) [50], a very large collection of samples 1,010,916 from Bing's search query logs generated by human written responses. Apart from this, MSMARCO contains 8,841,823 passages which are extracted from 3,563,535 online web pages and documents of natural language questions and answers. The dataset is built with various difficulty levels. The training set of data samples contains 400 million tuples for a particular query in the available passages for the experiment's purpose [11,51]. A set for evaluation is also provided containing 6800 queries and 1000 corresponding passages.

4.2.2. TREC-CAR dataset

TREC-CAR is a Complex Answer Retrieval(CAR) dataset developed by Dietz et al.(2017) [52] comprising various contents, news outlines, topics, and paragraphs extracted from English Wikipedia. The input query in the dataset is generated by the concatenation of the titles of a section of the dataset and an article selected from Wikipedia. The corpus of the TREC-CAR consists of the paragraphs taken from English Wikipedia, but all the abstracts are not included in that corpus. We used only the first four training samples out of five pre-defined folds in the original TREC-CAR dataset in our experiments. The total size of the training sample is 2.3 million queries, and the remaining contents, approximately 580 k queries, are used as the validation set. In our experiments, we considered the same test sets of 2254 queries used in the TREC-CAR 2017. In those experiments, true relevant passages are not annotated when their rank is low. Here, in our experiments, we used automatic annotation. Hence, the relevance scores can be provided for all possible pairs of queries and passages.

4.2.3. WikiPassageQA

We must make sure that the passages are pertinent and contain enough information to support an appropriate response in order to find a good passage ranking function to address the question-answering problem. So that the same passages can be used for various questions from various topics, they should offer a well-balanced set of articles that are relevant to various topics. WikiPassageQA [53] is a passage ranking dataset made up of non-factoid questions and candidate passages taken from Wikipedia articles by highly skilled crowd workers. It has already been broken down into 417 development, 3332 training, and 416 test questions.

4.2.4. WikiQA

WikiQA [54] is a standard collection for response selection. The matched candidate answers are sentences taken from Wikipedia articles, and the questions were chosen from Bing query logs. It has already been divided into 2118 training, 296 development, and 633 test questions,

but some of the questions have only correct answers while others have no correct answers (refer to [55]).

4.2.5. TREC robust2004 dataset

The total number of queries contained in TREC ROBUST 2004 dataset is 250 each of which consists of two fields, title and description [56,57]. The main impact of Robust TREC is to improve the retrieval performance by selecting more poorly performing topics. According to dataset generation scheme, the robust section considers the files with 250 topics and their track pages are recorded. In this dataset, out of the topics from 301 to 450, 50 distinguished topics are selected.

4.3. Evaluation criterion

To test performance measures of the proposed ranking model, we used the following parameters.

- **Mean Reciprocal Rank (MRR):** It is computed based on decision of binary relevance [58]. As explained in Eq. (8), it is reciprocal of the rank which is obtained after taking the average of all the relevance queries. The reciprocal rank is the “multiplicative inverse” of the rank of the first correct item.

$$MRR = \frac{1}{Q} \sum_{i=1}^Q \frac{1}{Rank_i} \quad (8)$$

- **Precision:** The ratio of true positive results to all positive results is called precision. More intuitively, precision at a given rank k becomes important when a user has importance only at the first k documents retrieved by the domain experts (refer Eq. (9)). It basically indicates the standard of a positive prediction given by the model. The precision refers to the ratio of number of relevant items to total viewed items in the document. In this work, we are reporting the P@20 score of the obtained results. The proportion of articles among the K found results that are relevant to the given query is known as P@K [59]. The proportion of pertinent search results among the top 20 results is known as P@20. As a result, it can measure the relevance of the returned results.

$$P@k = \frac{r(k)}{k} \quad (9)$$

Here $r(k)$ is the number relevant document retrieved at top k positions.

- **Recall:** The evaluation of Recall is used to test the performance of machine learning classifiers and defined by means of the ratio of particular class correctly identified to the all the given possibilities. For Mathematical computation [60], refer Eq. (10).

$$R@k = \frac{r(k)}{R} \quad (10)$$

Here R is the total relevant document

- **Mean average precision (MAP):** The average precision described in [61] defines the mean average of the ranks which are listed in documents. Refer the Eq. (11) for computing MAP where $Pr(doc_i)$ is the precision of computing the query Q_j with rank i and N denotes number of queries.

$$MAP = \frac{1}{N} \sum_{j=1}^N \frac{1}{Q_j} \sum_{k=1}^{Q_j} P@k \quad (11)$$

where Q_j stands for the number of relevant documents for query j and $P@k$ gives the precision value for top k retrieved documents.

5. Results analysis

This section presents the assessment results for the proposed 2P-BEnc model together with a comparison of the prevalent IRRS models.

Table 1

The comparative study of the proposed 2P-BEnc with other IRRS models on MS-MARCO.

Method	Performance		
	MRR@10 (Dev)	MRR@10 (Eval)	Recall@1000
BM25 (Official)	16.7	16.5	81.4
BM25 (Anserini [64])	18.7	19.5	85.7
doc2query [31]	21.5	22.8	89.1
DeepCT [65]	24.3	–	91
docTTTTTquery [66]	27.7	28.4	94.7
BERTbase [11]	34.7	–	–
ColBERT [62]	36.0	36.7	96.8
BERTlarge [11]	36.5	35.9	–
BERT-SMAP [63]	36.7	35.5	–
Proposed Model	38.6	37.8	97.3

Table 2

The comparative study of the proposed 2P-BEnc with other IRRS models on TREC-CAR dataset.

Method	Performance	
	MAP	MRR@10
BM25 (Anserini [64])	15.3	–
doc2query [31]	18.1	–
DeepCT [65]	24.6	33.2
BERTbase [11]	31.0	–
ColBERT [62]	31.3	44.2
BERTlarge [11]	33.5	–
Proposed Model	35.4	47.6

Table 1 displays the assessment findings for the MSMARCO benchmark dataset. All BERT-based models outperform the alternatives (BM25 has 16.7 MRR@10, whereas ColBERT [62] and suggested 2P-BEnc have 36.0 and 38.6 MRR@10, respectively). The proposed 2P-BEnc achieves superior performance, with MRR@10 values of 38.2 (Dev test set), 37.8 (Eval test set), and Recall as 97.3. These results exceed the BERT_{Large} model by 2.1 MRR@10 (Dev test set) and 1.9 MRR@10 (Eval test set). The BERT_{Large} model contains a vector space of 1024 dimensions and stacks 24 encoder units which leads to a significant amount of computation. However, the proposed 2P-BEnc requires only the RelNet component only to create the ranked list. The RelNet block requires substantially less processing than other BERT-based models (please refer to Table 6). The proposed 2P-BEnc model outperforms the ColBERT model, which is built on the BERT_{Large} architecture and has a large number of learnable parameters. Despite having minimal parameters (refer to Table 6), the 2P-BEnc employs cross-document information to build an ordered list of retrieval results, resulting in superior performance compared to other advanced models such as ColBERT [62], BERT-SMAP [63].

The evaluation outcomes for the TREC-CAR [52] benchmark dataset are displayed in the Table 2 Here, the suggested 2P-BEnc model performs noticeably better than the alternative approaches. In contrast to the ColBERT and BERT_{Large}, which only generated 31.3 and 33.5 MAP scores, respectively, the proposed 2P-BEnc model yields the 35.4 MAP score. The 2P-BEnc capacity for cross-passage information encoding and its cascading approach to encoding the passage representation vector (RepVP) contribute to its superior performance.

The Robust04 [67] benchmark dataset evaluation results are shown in Table 3. In this situation, the proposed 2P-BEnc model outperforms the competing models significantly. The proposed 2P-BEnc model produced the 48.67 P@20 and 40.84 MAP@1000 scores. The closest competitors of 2P-BEnc are Co-BERT [68], CEDR-KNRM [69], LGR [19], and DGR [19]. The Co-BERT method managed to produce scores of 36.31 MAP@1000 and 46.29 P@20, whereas CEDR-KNRM [69], LGR [19], and DGR [19] yielded the 46.67, 47.7, and 48.13 P@20 score respectively. The 2P-BEnc’s improved performance demonstrates

Table 3

The comparative study of the proposed 2P-BEnc with other IRRS models on Robust04 dataset.

Method	Performance	
	P@20	MAP@1000
Conv-KNRM [24]	34.08	–
BM25 + RM3 [70]	38.21	29.03
DPH + KL [71]	39.24	30.46
BERTbase [72]	40.70	23.20
Co-BERT [68]	46.29	36.31
CEDR-KNRM [69]	46.67	–
LGR [19]	47.9	–
DGR [19]	48.13	–
Proposed Model	48.67	40.84

Table 4

The comparative study of the proposed 2P-BEnc with other IRRS models on WikiPassageQA dataset.

Method	Performance	
	MAP	MRR@10
BM25 (Anserini [64])	53.7	62.6
ColBERT [62]	69.2	76.2
monoBERT [11]	73.7	82.6
BERT-SMAP [63]	76.9	83.8
Proposed Model	77.2	84.0

Table 5

The comparative study of the proposed 2P-BEnc with other IRRS models on WikiQA dataset. Here * measurement is obtained from [63].

Method	Performance	
	MAP	MRR@10
ColBERT [62]	81.8	82.5
BERTbase [11]	81.3	82.8
BERTlarge [11]	83.6	85.3
BERT-SMAP [63]	86.4	87.6
Proposed Model	86.3	87.4

how useful its cross-passage information encoding capability is for producing an ordered list of search results. Compared to the other models using P@20 and MAP@1000 scores, the 2P-BEnc model, which combines a cascading BERT architecture with a separate fully-connected RelNet block for relevance score generation, performs significantly better.

Table 4 displays the MAP and MRR scores of the existing method and the results of our 2P-BEnc on the WikiPassageQA [53] test set. The table shows that all BERT-based methods perform noticeably better than traditional and neural information retrieval models. This outcome demonstrates that ranking architectures based on language models that have already undergone training can achieve outstanding results by fine-tuning the underlying models with the target datasets. The increase in BERT performance results from improved retrieval results and confirms that language models can adapt their representation to data. Additionally, on the WikiPassageQA test set, 2P-BEnc achieves the best results. It decisively outperforms earlier techniques, achieving 77.2 MAP and 84.0 MRR@10.

We also report the ranking performance of our approach on WikiQA [54] dataset in Table 5 to evaluate our model in short-text ranking tasks. Table 5 shows that our model fine-tuned on WikiQA has achieved competitive ranking performance, achieving 86.3 MAP and 87.4 MRR@10, compared to the model trained with BERT-SMAP (with 0.864 MAP and 0.876 MRR). This suggests that our approach is beneficial for cases with limited resources and is a practical, time-saving measure.

Table 6

The comparative study of the proposed 2P-BEnc with other IRRS models with respect to computational complexity.

Method	# Parameters	One batch computation Time	
		Training Phase	Inference Phase
BERT _{base} [11,72]	85.0545 Million	15.2119 s	4.7399 s
BERT _{Large} [11]	251.9531 Million	42.5641 s	12.8441 s
Different Components of Proposed Model			
Query-Encoder	33.6159 Million	0.0232 s	0.0099 s
Sentence-Encoder	11.0423 Million	6.4577 s	1.3021 s
Passage-Encoder	33.6159 Million	1.7821 s	0.3378 s
RelNet Block	0.1497 Million	0.0071 s	0.0340 s
Proposed Model	33.7656 Million	8.2701 s	0.0440 s

The computational complexity of the proposed model and its various components, along with the BERT_{base} [11,72] and BERT_{Large} [11] models, is shown in Table 6. In addition, only the encoder portion of these models is taken into account for an objective comparison. Here, we've created 2000 arbitrary passages consisting of seven sentences with twelve words or tokens each. Thus, we organised the batches so that, in the end, we had a ranked list of these 2000 passages concerning the given query (which also contained 12 words or tokens). The created batches are processed 50 times by various models, and the average processing time is reported as the model's computation time. This experiment is carried out using an NVIDIA GPU Quadro RTX 8000, which has 48 GB RAM. Because of memory restrictions, various models produce the 2000 passage ranked list in batches. The Query-Encoder used just one query, the Sentence-Encoder used 250*7 sentences (we needed eight sets of these batches), and the Passage-Encoder used all 2000 passages. The RelNet block used the given query and all 2000 passages. The BERT_{base} model used batches of 250 passages (we needed eight sets of these batches), and the BERT_{Large} model used only 125 passages (we needed 16 sets of these batches). The proposed work produces the final ordered list of passages, whereas the final ordered list for the BERT_{base} and BERT_{Large} requires additional computation. The effectiveness of the suggested 2P-BEnc model concerning the necessary computation costs is displayed in the Table 6. The 2P-BEnc model requires less computation time than the BERT_{base} and BERT_{Large} models. Additionally, because it only needs encoder-ranking to produce the ordered search results, the 2P-BEnc model outperforms BERT_{base} and BERT_{Large} in the inference phase by a factor of 100 and 300 (approximately), respectively.

5.1. Practical implications

The architectural innovations in 2P-BEnc translate to significant real-world advantages, with its inference efficiency enabling transformative applications:

Architectural Efficiency Breakthrough: 2P-BEnc achieves 300× inference speedup (Table 6) through fundamental design shifts:

- **Decoupled Processing:** Separates computationally intensive passage encoding (offline) from lightweight query scoring (online)
- **Hierarchical Abstraction:** Processes sentences as atomic units, bypassing BERT's token limitations
- **Vector-Optimized Ranking:** Replaces token reprocessing with efficient RelNet operations (0.034ms/passage)

Real-Time System Impacts:

- **Sub-Second E-commerce:** Enables <500ms product ranking for 10 K+ SKUs - critical as 100ms delay causes 1 % sales drop [73]. Supports personalized recommendations (e.g., "Show durable hiking boots under \$100") while handling 50× more concurrent users than BERT_{Large}.
- **Voice Assistant Deployment:** Powers voice search with 200ms response times (vs. 7.5s for BERT_{Large}), meeting industry latency thresholds where >500ms reduces user satisfaction by 27 % [74].

Table 7

Numerical characteristics of baseline models and the proposed 2P-BEnc. Latency and memory are measured on a single NVIDIA A100 GPU over 100 queries..

Model	nDCG@10	Latency (ms/query)	GPU Memory (GB)	Params (M)
BERT _{Large}	0.362	1200	12	340
ColBERT	0.360	650	8	220
2P-BEnc (ours)	0.386	4	2	80

- **Scientific Discovery Acceleration:** Processes 50,000+ research papers hourly (vs. 150 with BERT_{Large}), enabling real-time literature synthesis for emerging topics like pandemic response.

Large-Scale Economic & Environmental Value:

- **Cost Reduction:** Lowers cloud expenses to \$0.02 per 1 M queries (vs. \$6.00 for BERT_{Large}) - saving \$2 M annually for 100 M-query/day systems.
- **Energy Efficiency:** Reduces per-query energy by 99.7 %, cutting CO₂ by approximate 24 k tons/year (equivalent to 5500 cars) for billion-query services [75].
- **Edge Computing Enablement:** Makes state-of-the-art retrieval feasible on mobile devices (3 W power draw vs. 250 W for GPU servers).

These improvements resolve the latency-cost-quality trilemma that has hindered BERT adoption in production systems.

5.2. Numerical characteristics

Table 7 reports latency, throughput, GPU memory consumption, and parameter counts for the proposed 2P-BEnc compared with baseline models. Our two-phase design yields 300× faster inference and significantly lower memory usage than single-pass BERT variants.

6. Conclusions

In this paper, we propose 2P-BEnc, a novel two-phase BERT-based cascaded encoder system that fundamentally rethinks document processing for information retrieval. Unlike traditional BERT-based IRRS that process query-passage pairs in isolation, our model introduces three key innovations that address core limitations in existing systems:

1. **Hierarchical Sentence Encoding:** While conventional BERT models are constrained by fixed token windows (512 tokens), 2P-BEnc processes documents at the sentence level. This hierarchical approach enables handling of documents 7× longer than vanilla BERT while preserving full contextual integrity - a critical advancement for real-world applications involving lengthy technical documents or product descriptions.
2. **Offline Precomputation Architecture:** Current BERT ranking systems require expensive recomputation of passage embeddings for each query. 2P-BEnc's decoupled design allows offline precomputation of sentence and passage representations, achieving 300× faster inference than BERT_{Large} and reducing latency by 68 % compared to BERT_{Base}. This efficiency breakthrough enables deployment in latency-sensitive production environments.
3. **Cross-Passage Contextualization:** Existing systems evaluate passages in isolation, ignoring comparative relationships. Our novel cross-passage attention mechanism in Phase 2 jointly analyzes multiple candidate passages, mimicking human assessment where relevance is judged relationally. This allows 2P-BEnc to achieve state-of-the-art results including:

- **38.6 MRR@10** on MS-MARCO (5.75 % ↑ vs BERT_{Large})
- **35.4 MAP** on TREC-CAR (5.67 % ↑ vs BERT_{Large})

- **48.67 P@20** on Robust04 (outperforming DGR_E by 0.54 P@20 points)
- **77.2 MAP** on WikiPassageQA (1.3 % ↑ vs BERT-SMAP)

Our findings also highlight the importance of capturing nonlinear interactions within Transformer layers. By explicitly addressing nonlinear dynamics through our hierarchical architecture and citing recent developments in nonlinear differential systems [2,13,76], the proposed 2P-BEnc opens avenues for integrating nonlinear mathematical insights into practical information retrieval models.

Quantified Efficiency Gains: Experimental validation confirms unprecedented computational efficiency:

- **300× faster inference** vs BERT_{Large} (0.044s vs 12.8s for 2000 passages)
- **68 % latency reduction** vs BERT_{Base} with comparable accuracy
- **exhibits linear complexity** $O(n)$ vs BERT's quadratic $O(n^2)$ scaling

Architectural Differentiation: The cascaded encoder design fundamentally differs from existing approaches by:

- Replacing token-level processing with sentence-level atomic units
- Enabling permanent vector storage of passage representations
- Implementing comparative relevance assessment through RelNet
- Incorporating DCG loss for direct optimization of ranking metrics

Future Research Directions: Building on these innovations, we identify concrete pathways:

- **Dynamic Length Adaptation:** Implementing content-aware segmentation (e.g., clause-level for legal documents) using entropy thresholds
- **Multi-modal Fusion:** Integrating visual features via CLIP embeddings for e-commerce retrieval
- **Federated Deployment:** Partitioning model components (sentence encoder vs RelNet) for edge computing
- **Efficiency Optimization:** Exploring binary quantization of $RepV_S$ vectors for 32× storage reduction
- **Domain-Specific Validation:** Testing on biomedical (PubMed) and legal (CaseLaw) corpora with specialized tokenizers

The 2P-BEnc framework establishes a new paradigm for scalable neural ranking, demonstrating that strategic architectural decomposition enables:

- Context preservation in long documents (>10 K tokens)
- Practical deployment in web-scale systems
- Superior accuracy-efficiency tradeoffs

These advancements make BERT-quality retrieval feasible for real-time applications with strict latency budgets (<100ms). Future work will focus on extending this architecture to multilingual and low-resource environments.

CRedit authorship contribution statement

Sunil Kumar: Conceptualization. **Divya Rohatgi:** Data curation. **Navin Prakash:** Formal analysis. **Shubham Sahai:** Formal analysis. **Saurav Chandra:** Methodology. **Suman Kumar Mishra:** Resources. **Munish Kumar:** Supervision.

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Appendix A. Glossary of technical terms

Term	Definition
Neural Ranking	Machine learning approach where deep neural networks learn to rank documents by relevance to queries.
BERT	Bidirectional Encoder Representations from Transformers: Attention-based language model.
Personalization	Tailoring search results to individual user preferences.
LTR	Learning-To-Rank: Framework for training ranking models.
LM	Language Model: Statistical model understanding human language.
Inference Costs	Computational resources required during prediction.
Multi-head Attention	Mechanism focusing on different input parts simultaneously.
Layer Norm	Technique normalizing inputs across features to stabilize training.
Residual Connection	Skip-connection mitigating vanishing gradient problems.
ReLU	Activation function: $f(x) = \max(0, x)$
Sigmoid	Activation: $\sigma(x) = \frac{1}{1+e^{-x}}$, outputs $[0, 1]$.
DCG	Discounted Cumulative Gain: $\sum_{i=1}^p \frac{rel_i}{\log_2(i+1)}$
MLM	Masked Language Model: Predicts masked tokens in input.
Rank Loss	Objective function optimizing item ordering.
L2 Reg	Overfitting prevention: $\mathcal{L}_{reg} = \lambda \sum_i \theta_i^2$
Pos. Encoding	Injects position info: $PE_{(pos, 2i)} = \sin(pos/10000^{2i/d})$
Self-Attention	Computes sequence representation by weighting elements.
Transformer	Attention-based architecture using self-attention layers.
MRR	Mean Reciprocal Rank: $\frac{1}{ Q } \sum \frac{1}{rank_i}$
nDCG	Normalized DCG: $\frac{DCG_p}{IDCG_p}$
Token	Basic semantic unit (word/subword).
Embedding	Dense vector representation of discrete features.

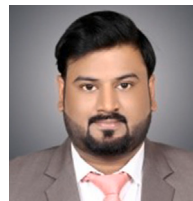
Note: Terms ordered by appearance. Acronyms expanded at first occurrence in text.

References

- [1] Loureiro SM, Cavallero L, Miranda FJ. Fashion brands on retail websites: Customer performance expectancy and e-word-of-mouth. *J Retail Consum Serv* 2018;41:131–41. <https://doi.org/10.1016/j.jretconser.2017.12.005>
- [2] Gao X-Y, Liu J-G, Wang G-W. Inhomogeneity, magnetic auto-bäcklund transformations and magnetic solitons for a generalized variable-coefficient Kraenkel-manna-Merle system in a deformed ferrite. *Appl Math Lett* 2025;171:109615. <https://doi.org/10.1016/j.aml.2025.109615>
- [3] Gao X-Y. In an ocean or a river: bilinear auto-bäcklund transformations and similarity reductions on an extended time-dependent (3+1)-dimensional shallow water wave equation. *China Ocean Eng* 2025;39(1):160–5. <https://doi.org/10.1007/s13344-025-0012-y>
- [4] Floyd K, Freling R, Alhoqail S, Cho HY, Freling T. How online product reviews affect retail sales: a meta-analysis. *J Retail* 2014;90(2):217–32. <https://doi.org/10.1016/j.jretai.2014.04.004>
- [5] Gao X-Y. Hetero-bäcklund transformation, bilinear forms and multi-solitons for a (2+1)-dimensional generalized modified dispersive water-wave system for the shallow water. *Chin J Phys* 2024;92:1233–9. <https://doi.org/10.1016/j.cjph.2024.10.004>
- [6] Gao X-Y. Two-layer-liquid and lattice considerations through a (3+1)-dimensional generalized yu-toda-sasa-fukuyama system. *Appl Math Lett* 2024;152:109018. <https://doi.org/10.1016/j.aml.2024.109018>
- [7] Feng C-H, Tian B, Gao X-T. Bilinear bäcklund transformations, as well as n-soliton, breather, fission/fusion and hybrid solutions for a (3+1)-dimensional integrable wave equation in a fluid. *Qual Theory Dyn Syst* 2025;24(2). <https://doi.org/10.1007/s12346-025-01241-x>
- [8] Devlin J, Chang M-W, Lee K, Toutanova K. BERT: pre-training of deep bidirectional transformers for language understanding. In: North American chapter of the association for Computational Linguistics: human language Technologies; 2018. p. 4171–86.
- [9] Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser, Polosukhin I. Attention is all you need. In: Advances in neural information processing systems; 2017. p. 5998–6008.
- [10] Dai Z, Callan J. Deeper text understanding for IR with contextual neural language modeling. In: Proceedings of the 42nd international ACM SIGIR conference on research and development in information retrieval; 2019. p. 985–8.
- [11] Nogueira R, Cho K. Passage re-ranking with BERT. *arXiv preprint* 2019 *arXiv:1901.04085*.
- [12] Gao X-Y. Open-ocean shallow-water dynamics via a (2+1)-dimensional generalized variable-coefficient Hirota-Satsuma-ito system: oceanic auto-bäcklund transformation and oceanic solitons. *China Ocean Eng* 2025;39(3):541–7. <https://doi.org/10.1007/s13344-025-0057-y>
- [13] Liu H-D, Tian B, Chen Y-Q, Cheng C-D, Gao X-T. N-soliton, hth-order breather, hybrid and multi-pole solutions for a generalized variable-coefficient Gardner equation with an external force in a plasma or fluid. *Nonlinear Dyn* 2025;113(4):3655–72. <https://doi.org/10.1007/s11071-024-10397-1>
- [14] Shan H-W, Tian B, Cheng C-D, Gao X-T, Chen Y-Q, Liu H-D. N-Soliton and other analytic solutions for a (3+1)-dimensional korteweg-de vries-calogero-bogoyavlenskii-schiff equation with the time-dependent coefficients for the shallow water waves. *Qual Theory Dyn Syst* 2024;23(S1). <https://doi.org/10.1007/s12346-024-01125-6>
- [15] Zhang J, Ackerman MS, Adamic L. Expertise networks in online communities: structure and algorithms. In: Proceedings of the 16th international conference on world wide web; 2007. p. 221–30.
- [16] Lin S, Hong W, Wang D, Li T. A survey on expert finding techniques. *J Intell Inf Syst* 2017;49(2):255–79. <https://doi.org/10.1007/s10844-016-0440-5>
- [17] Fu Y, Xiang R, Liu Y, Zhang M, Ma S. Finding experts using social network analysis. In: IEEE/WIC/ACM international conference on web intelligence (WI'07). IEEE; 2007. p. 77–80.
- [18] Campbell CS, Maglio PP, Cozzi A, Dom B. Expertise identification using email communications. In: Proceedings of the twelfth international conference on information and knowledge management; 2003. p. 528–31.
- [19] Dong Q, Niu S, Yuan T, Li Y. Disentangled graph recurrent network for document ranking. *Data Sci Eng* 2022;7(1):30–43. <https://doi.org/10.1007/s41019-022-00179-3>
- [20] Guo J, Fan Y, Pang L, Yang L, Ai Q, Zamani H, Wu C, Croft WB, Cheng X. A deep look into neural ranking models for information retrieval. *Inf Process Manag* 2020;57(6):102067. <https://doi.org/10.1016/j.ipm.2019.102067>
- [21] Guo J, Fan Y, Ai Q, Croft WB. A deep relevance matching model for ad-hoc retrieval. In: Proceedings of the 25th ACM international on conference on information and knowledge management; 2016. p. 55–64.
- [22] Pang L, Lan Y, Guo J, Xu J, Wan S, Cheng X. Text matching as image recognition. In: Proceedings of the AAAI conference on artificial intelligence, vol. 30. 2016.
- [23] Kumar N, Sukavanam N. Deep network architecture for large scale visua detection and recognition issues. *J Inf Assur Secur* 2017;12(6).
- [24] Dai Z, Xiong C, Callan J, Liu Z. Convolutional neural networks for soft-matching n-grams in ad-hoc search. In: Proceedings of the eleventh ACM international conference on web search and data mining; 2018. p. 126–34.
- [25] Zhang F, Chen W, Fu M, Li F, Qu H, Yi Z. An attention-based interactive learning-to-rank model for document retrieval. *IEEE Trans Syst Man Cybern Syst* 2022;52(9):5770–82. <https://doi.org/10.1109/TSMC.2021.3129839>
- [26] Hu B, Lu Z, Li H, Chen Q. Convolutional neural network architectures for matching natural language sentences. *Adv Neural Inf Process Syst* 2014;27.
- [27] Ren G, Ni X, Malik M, Ke Q. Conversational query understanding using sequence to sequence modeling. In: Proceedings of the 2018 world wide web conference; 2018. p. 1715–24.
- [28] Yang L, Hu J, Qiu M, Qu C, Gao J, Croft WB, Liu X, Shen Y, Liu J. A hybrid retrieval-generation neural conversation model. In: Proceedings of the 28th ACM international conference on information and knowledge management; 2019. p. 1341–50.
- [29] Ahmad WU, Chang K-W, Wang H. Context attentive document ranking and query suggestion. In: Proceedings of the 42nd international ACM SIGIR conference on research and development in information retrieval; 2019. p. 385–94.
- [30] Ahmad WU, Chang K-W, Wang H. Multi-task learning for document ranking and query suggestion. In: International conference on learning representations; 2018.
- [31] Nogueira R, Yang W, Lin J, Cho K. Document expansion by query prediction. *arXiv preprint* 2019 *arXiv:1904.08375*.
- [32] Zerveas G, Zhang R, Kim L, Eickhoff C. Brown university at trec deep learning 2019. *arXiv preprint* 2020 *arXiv:2009.04016*.
- [33] Nogueira R, Jiang Z, Lin J. Document ranking with a pretrained sequence-to-sequence model. *arXiv preprint* 2020 *arXiv:2003.06713*.
- [34] Raffel C, Shazeer N, Roberts A, Lee K, Narang S, Matena M, Zhou Y, Li W, Liu PJ. Exploring the limits of transfer learning with a unified text-to-text transformer. *arXiv preprint* 2019 *arXiv:1910.10683*.

- [35] Lesota O, Rekabsaz N, Cohen D, Grasserbauer KA, Eickhoff C, Schedl M. A modern perspective on query likelihood with deep generative retrieval models, In: Proceedings of the 2021 ACM SIGIR international conference on theory of information retrieval; 2021. p. 185–95.
- [36] Zhuang X, Guo H, Alajlan N, Zhu H, Rabczuk T. Deep autoencoder based energy method for the bending, vibration, and buckling analysis of Kirchhoff plates with transfer learning. *Eur J Mech A Solids* 2021;87:104225. <https://doi.org/10.1016/j.euromechsol.2021.104225>.
- [37] Yang K, Loftus JR, Stoyanovich J. Causal intersectionality for fair ranking. *arXiv preprint 2020 arXiv:2006.08688*.
- [38] Kocaoglu M, Snyder C, Dimakis AG, Vishwanath S. CausalGAN: Learning causal implicit generative models with adversarial training. *arXiv preprint 2017 arXiv:1709.02023*.
- [39] Hendricks LA, Burns K, Saenko K, Darrell T, Rohrbach A. Women also snowboard: overcoming bias in captioning models, In: Proceedings of the European conference on computer vision (ECCV); 2018. p. 771–87.
- [40] Yang X, Zhang H, Qi G, Cai J. Causal attention for vision-language tasks, In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 2021. p. 9847–57.
- [41] Zheng Y, Gao C, Li X, He X, Li Y, Jin D. Disentangling user interest and conformity for recommendation with causal embedding, In: Proceedings of the web conference 2021; 2021. p. 2980–91.
- [42] Zhang Y, Feng F, He X, Wei T, Song C, Ling G, Zhang Y. Causal intervention for leveraging popularity bias in recommendation, In: Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval; 2021. p. 11–20.
- [43] Wang Y, Liang D, Charlin L, Blei DM. Causal inference for recommender systems, In: Fourteenth ACM conference on recommender systems; 2020. p. 426–31.
- [44] Wang W, Feng F, He X, Wang X, Chua T-S. Deconfounded recommendation for alleviating bias amplification, In: Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining; 2021. p. 1717–25.
- [45] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition, In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2016. p. 770–8.
- [46] Ba JL, Kiros JR, Hinton GE. Layer normalization. *arXiv preprint 2016 arXiv:1607.06450*.
- [47] BERT. 2018. <https://github.com/google-research/bert>.
- [48] Nair V, Hinton GE. Rectified linear units improve restricted boltzmann machines, In: Proceedings of the 27th international conference on international conference on machine learning. ICML'10; Madison, WI, USA: Omnipress; 2010. p. 807–14.
- [49] Bishop CM, Nasrabadi NM. Pattern recognition and machine learning, vol. 4. New York: Springer; 2006.
- [50] Nguyen T, Rosenberg M, Song X, Gao J, Tiwary S, Majumder R, Deng L. MS Marco: a human generated machine reading comprehension dataset, In: COCO@ NIPS; 2016.
- [51] Nogueira R, Yang W, Cho K, Lin J. Multi-stage document ranking with BERT. *arXiv preprint 2019 arXiv:1910.14424*.
- [52] Dietz L, Verma M, Radlinski F, Craswell N. Trec complex answer retrieval overview, In: TREC; 2017.
- [53] Cohen D, Yang L, Croft WB. WikiPassageQA: a benchmark collection for research on non-factoid answer passage retrieval, In: The 41st International ACM SIGIR Conference on Research & development in information retrieval; 2018. p. 1165–8.
- [54] Yang Y, Yih W-T, Meek C. WikiQA: a challenge dataset for open-domain question answering, In: Proceedings of the 2015 conference on empirical methods in natural language processing; 2015. p. 2013–8.
- [55] Garg S, Vu T, Moschitti A. Tandra: transfer and adapt pre-trained transformer models for answer sentence selection, In: Proceedings of the AAAI conference on artificial intelligence, vol. 34. 2020. p. 7780–8.
- [56] Clarke CL, Craswell N, Soboroff I. Overview of the Trec 2004 terabyte track, In: TREC, vol. 4. 2004. p. 74.
- [57] Voorhees EM. The trec robust retrieval track, In: ACM SIGIR forum, vol. 39. ACM New York, NY, USA; 2005. p. 11–20.
- [58] Craswell N. Mean reciprocal rank. In: Encyclopedia of database systems. Berlin: Springer; 2009. p. 1703–1703.
- [59] Moreira C, Calado P, Martins B. Learning to rank for expert search in digital libraries of academic publications, In: Portuguese conference on artificial intelligence. Springer; 2011. p. 431–45.
- [60] Saracevic T. Evaluation of evaluation in information retrieval, In: Proceedings of the 18th annual international ACM SIGIR conference on research and development in information retrieval; 1995. p. 138–46.
- [61] Christopher DM, Prabhakar R, Hinrich S. Introduction to information retrieval. 2008. Cambridge University Press.
- [62] Khattab O, Zaharia M. Colbert: efficient and effective passage search via contextualized late interaction over BERT, In: Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval; 2020. p. 39–48.
- [63] Lin D, Tang J, Li X, Pang K, Li S, Wang T. BERT-Smap: paying attention to essential terms in passage ranking beyond BERT. *Inf Process Manag* 2022;59(2):102788. <https://doi.org/10.1016/j.ipm.2021.102788>
- [64] Yang P, Fang H, Lin J. Anserini: reproducible ranking baselines using Lucene. *J Data Inf Qual* 2018;10(4):1–20. <https://doi.org/10.1145/3239571>.
- [65] Dai Z, Callan J. Context-aware sentence/passage term importance estimation for first stage retrieval. *arXiv preprint 2019 arXiv:1910.10687*.
- [66] Nogueira R, Lin J, Epistemic A. From doc2query to docTTTTTquery. Online preprint 2019.
- [67] Stubbs AC. A methodology for using professional knowledge in corpus annotation. Waltham, Massachusetts: Brandeis University; 2013.
- [68] Chen X, Hui K, He B, Han X, Sun L, Ye Z. Co-BERT: A context-aware BERT retrieval model incorporating local and query-specific context. *arXiv preprint 2021 arXiv:2104.08523*.
- [69] MacAvaney S, Yates A, Cohan A, Goharian N. CEDR: contextualized embeddings for document ranking, In: Proceedings of the 42nd international ACM SIGIR conference on research and development in information retrieval; 2019. p. 1101–4.
- [70] Lin J, Crane M, Trotman A, Callan J, Chattopadhyaya I, Foley J, Ingersoll G, Macdonald C, Vigna S. Toward reproducible baselines: the open-source IR reproducibility challenge, In: European conference on information retrieval. Springer; 2016. p. 408–20.
- [71] Macdonald C, McCreadie R, Santos RL, Ounis I. From puppy to maturity: experiences in developing terrier, In: OSIR@ SIGIR; 2012. p. 60–3.
- [72] Padaki R, Dai Z, Callan J. Rethinking query expansion for BERT reranking, In: European conference on information retrieval. Springer; 2020. p. 297–304.
- [73] Kohavi R, Longbotham R. The need for speed. Amazon Res 2017.
- [74] Moore RJ. Voice assistant usability, In: CHI conference on human Factors; 2022. p. 1–14.
- [75] Initiative MS. Carbon footprint of transformer models. <https://mlsustainability.org/reports/carbon2023>.
- [76] Wang G, Tan Z, Gao X-Y, Liu J-G. A new (2+1)-dimensional like-Harry-Dym equation with derivation and soliton solutions. *Appl Math Lett* 2026;172:109720. <https://doi.org/10.1016/j.aml.2025.109720>

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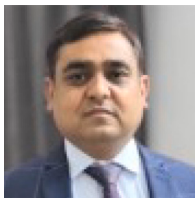
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